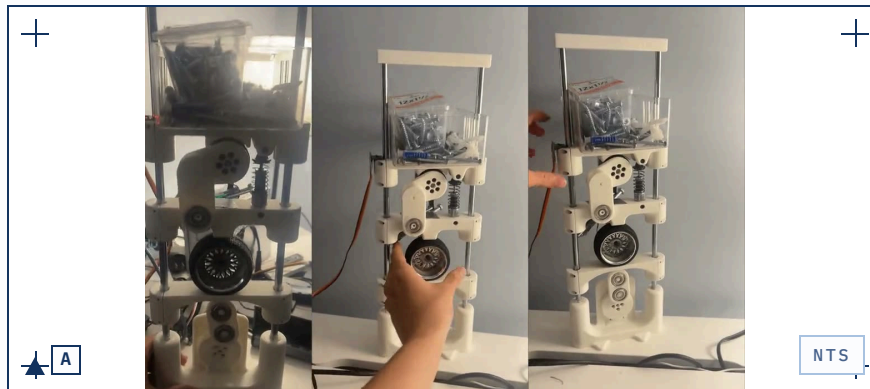


Active Suspension: Phase 2

Building the phase 1 model into a physical rig. Friction and motor cogging overdamped the system, so skyhook could not run, but the build taught me how to design around imperfect hardware.

BUILT IT FOR REAL



THE RIG RUNNING / 3 HZ ROAD, DROP TEST, PUSH DOWN AND PULL UP

01



FINAL RIG / MODELLED IN SOLIDWORKS

02



THE ASSEMBLED RIG

A GOAL

Phase 1 proved skyhook control in simulation. Phase 2 was meant to build that quarter-car model in hardware and run the controller for real. I built the rig. The controller never ran: friction and motor cogging overdamped the system, and skyhook is undefined on an already-overdamped plant. This page is the honest version of that, because the build is where I learned the most.

B METHOD

I needed every element of the phase 1 model as a physical part, constrained to move only vertically on linear rails. The piece with no obvious answer was the road: I turned a servo through a crank and pushrod to convert rotation into vertical travel, and used the same trick for the skyhook actuator.

C OUTCOME

I did not get skyhook running on hardware, and I am not going to pretend otherwise. What I got was the part that does not show up in a simulation: where the model quietly assumes an ideal, and how to design around hardware that refuses to be ideal. The full parts list, measurements, and derivations are in the Phase 2 writeup.

TOOLS

SolidWorks

3D Printing

Direct-Drive Motor

IMU

Hardware

DRAWN BY

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SHEET

01 / 06

DWG NO.

BTP-01

DIMENSIONS IN MM / TOLERANCES ± 0.5 UNLESS NOTED / DO NOT SCALE DRAWING

DETAILS

benjamintranpugh.com/projects/active-suspension-phase-2/ →

THIRD ANGLE
PROJECTION

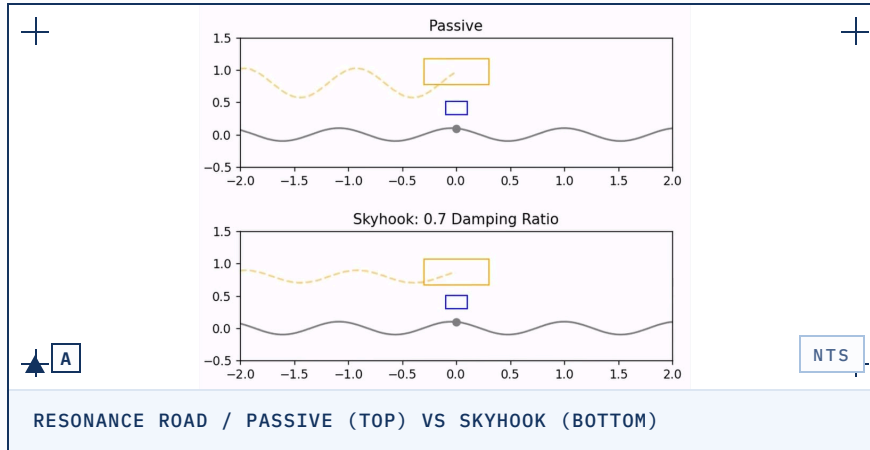
NOT FOR CONSTRUCTION

Active Suspension: Phase 1

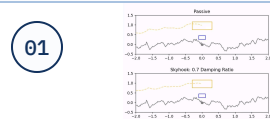
Skyhook control on a quarter-car model, cutting RMS body acceleration by 9.6% on a worn road and 57.1% at resonance.

RMS ACCEL: -9.6% TO -57.1%

ACTUATOR: 663 N PEAK

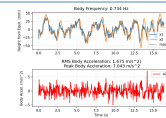


01



CLASS-C ROAD / PASSIVE (TOP)
VS SKYHOOK (BOTTOM)

02



CLASS-C ROAD / PASSIVE / RMS
1.675 M/S²

A GOAL

I built a quarter-car suspension model from scratch to answer one question: how much ride comfort can an actuator buy you over a passive spring and damper? Comfort is measured as RMS body acceleration, so that is the number I optimized and the number I report.

B METHOD

Most derivations I found waved away gravity, so I worked the equations of motion out from first principles instead. Writing the spring forces in absolute coordinates, then substituting each mass's static equilibrium position, makes the mg terms cancel exactly. That is the reason gravity drops out, and it also fixes what the coordinates mean: every displacement below is measured from static equilibrium, not from the ground.

C OUTCOME

This page is the summary. I wrote a full Phase 1 writeup that carries every step at length: the equations of motion derived line by line, why each parameter was chosen and checked against Gillespie, the complete ISO 8608 and variance derivations, and all eight benchmark cases with their plots. If you want to see how I actually work through a problem, read that.

TOOLS Python Control Systems Vehicle Dynamics ODEs ISO 8608

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SHEET

02 / 06

DWG NO.

BTP-02

DIMENSIONS IN MM / TOLERANCES ± 0.5 UNLESS NOTED / DO NOT SCALE DRAWING

DETAILS

benjamintranpugh.com/projects/active-suspension/ →



THIRD ANGLE
PROJECTION

NOT FOR CONSTRUCTION

Robot Arm 2.0

A 4-DOF robot arm with full inverse kinematics, browser-based control, and positional memory.

DOF: 4

IK: FULL

MCU: ESP32



THE BUILT ARM



01



SOLIDWORKS ASSEMBLY

02



CAD ORBIT / SOLIDWORKS

A GOAL

The goal: a functioning 4-DOF robot arm with full inverse kinematics, smooth motion, easy controls, and working positional memory, a major step up from my earlier 3-DOF build.

B METHOD

I started by researching the math behind inverse kinematics and robotic motion, then modeled the full arm in SolidWorks around accurate models of off-the-shelf components.

C OUTCOME

A working arm with full inverse kinematics, smooth controlled motion, and positional memory, controllable from any device on the network.

TOOLS

SolidWorks

FEA

C++

Python

MATLAB

ESP32

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SHEET

03 / 06

DWG NO.

BTP-03

DIMENSIONS IN MM / TOLERANCES ± 0.5 UNLESS NOTED / DO NOT SCALE DRAWING

DETAILS

benjamintranpugh.com/projects/robot-arm-2/ →THIRD ANGLE
PROJECTION

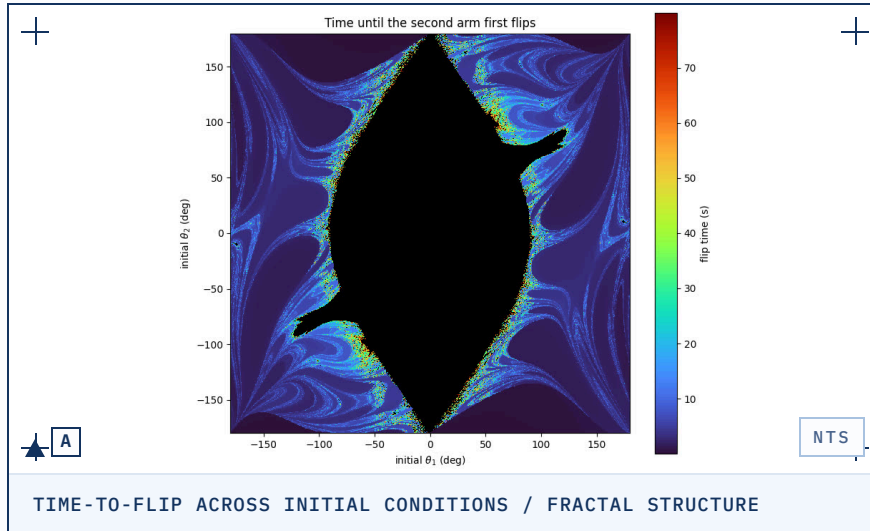
NOT FOR CONSTRUCTION

Double Pendulum

Deriving the equations of motion from the Euler–Lagrange equation and simulating chaos in Python.

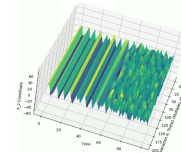
EOM: LAGRANGE

SOLVER: NUMERIC



TIME-TO-FLIP ACROSS INITIAL CONDITIONS / FRACTAL STRUCTURE

01



200 RUNS, $\Delta\theta_0 = 1e-9$ RAD /
DIVERGENCE OVER TIME

02



EULER-LAGRANGE DERIVATION,
TYPESET IN LATEX

A GOAL

The double pendulum is one of the simplest systems that produces chaos. I derived its equations of motion from the Euler–Lagrange equation and solved the resulting non-linear ODEs numerically in Python to explore how chaotic it really is.

B METHOD

- Derived the equations of motion with Lagrangian mechanics (full derivation typeset in LaTeX)
- Solved the coupled non-linear ODEs numerically in Python
- Swept initial conditions to map flip times and divergence behavior

C OUTCOME

To put into perspective how small $1e-9$ rad is, if you shined two lasers at the moon separated by $1e-9$ rad, the distance between them would be just over a foot.

TOOLS

Python

Lagrangian Mechanics

ODEs

LaTeX

Linear Algebra

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SHEET

04 / 06

DWG NO.

BTP-04

DIMENSIONS IN MM / TOLERANCES ± 0.5 UNLESS NOTED / DO NOT SCALE DRAWING

DETAILS

benjamintranpugh.com/projects/double-pendulum/ →THIRD ANGLE
PROJECTION

NOT FOR CONSTRUCTION

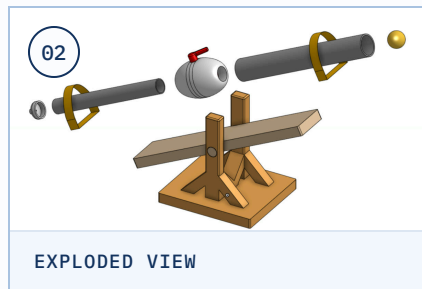
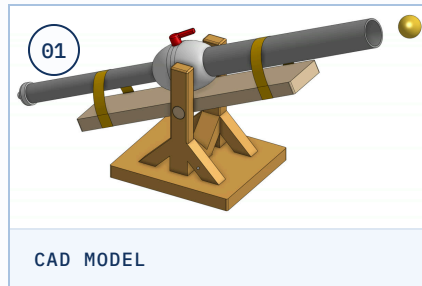
Air Cannon

A pneumatic launcher on needed for a 10 m target. It hit 60+ m.

TYPE:PNEUMATIC

SPEC:10M

ACTUAL:60M+



A GOAL

The assignment was a tennis-ball launcher with a maximum range of 10 meters, any more was supposed to be difficult. We decided that wasn't enough and built a pneumatic cannon that launches projectiles over 60 meters.

B METHOD

The cannon is built from PVC piping, wood, duct tape, and epoxy, pressurized with a bicycle tire pump. Getting there meant pressure research to size the chamber, CAD models in Fusion 360 and Onshape, and a lot of hands-on work, such as hand sawing, PVC solvent welding, and epoxy joints.

C OUTCOME

60+ meters of range from a hand-pumped chamber, and a hard lesson in the difference between a CAD model and a thing you can actually manufacture.

TOOLS Fusion 360 Onshape Pneumatics Fabrication

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SHEET

05 / 06

DWG NO.

BTP-05

DIMENSIONS IN MM / TOLERANCES ± 0.5 UNLESS NOTED / DO NOT SCALE DRAWING

DETAILS

benjamintranpugh.com/projects/air-cannon/ →

THIRD ANGLE
PROJECTION

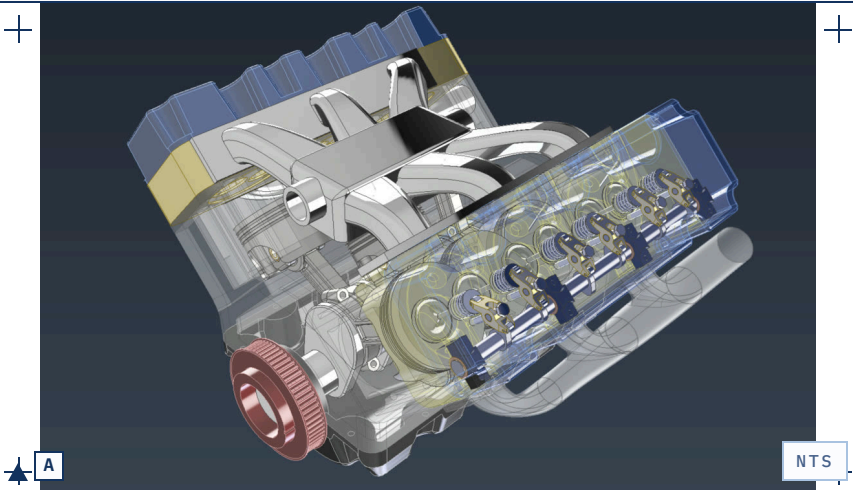
NOT FOR CONSTRUCTION

V6 Engine Model

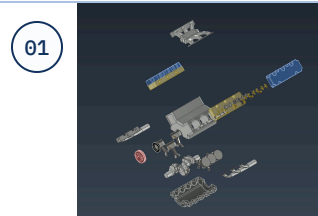
A detailed V6 engine assembly modeled from real engineering drawings in Autodesk Inventor.

CONFIG:V6

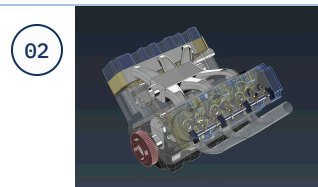
RENDER:INVENTOR STUDIO



FULL ASSEMBLY / VALVETRAIN AND CRANK VISIBLE THROUGH TRANSPARENT HEAD



EXPLODED VIEW



ORBIT RENDER / ANIMATED IN INVENTOR STUDIO

A GOAL

While researching how different internal combustion engines work, I found engineering drawings of V6 components, so I modeled the whole engine. I had just gotten access to Autodesk Inventor, and this became my crash course in it.

B METHOD

Working from the component drawings, I modeled each part and built up the full assembly, then used Inventor Studio for rendering and animation of the moving engine.

C OUTCOME

A complete, animated V6 model, and a much deeper understanding of what every component in an engine actually does. The orbit render below was animated in Inventor Studio.

TOOLS Autodesk Inventor Rendering Animation Large Assemblies

DRAWN BY

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SHEET

06 / 06

DWG NO.

BTP-06

DIMENSIONS IN MM / TOLERANCES ± 0.5 UNLESS NOTED / DO NOT SCALE DRAWING

DETAILS

benjamintranpugh.com/projects/v6-engine/ →

THIRD ANGLE
PROJECTION

NOT FOR CONSTRUCTION